



Warith Al-Anbiyaa University
College of Medicine
Department Microbiology-
Bacteriology



Unit 3: Lecture: Microbiology:

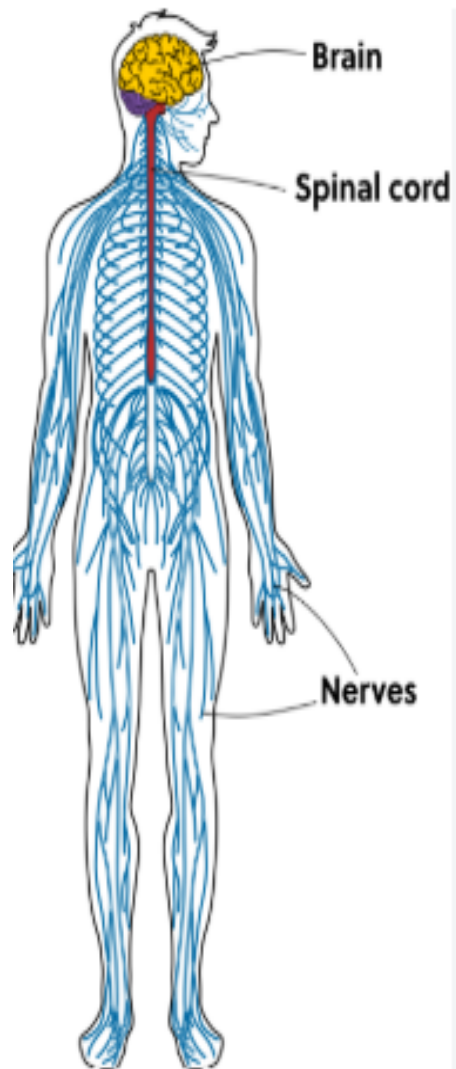
Infectious diseases affecting the peripheral nervous system,
focusing on shingles

Ass. Prof. Dr Nisreen Jawad Kadhim

Learing objectives

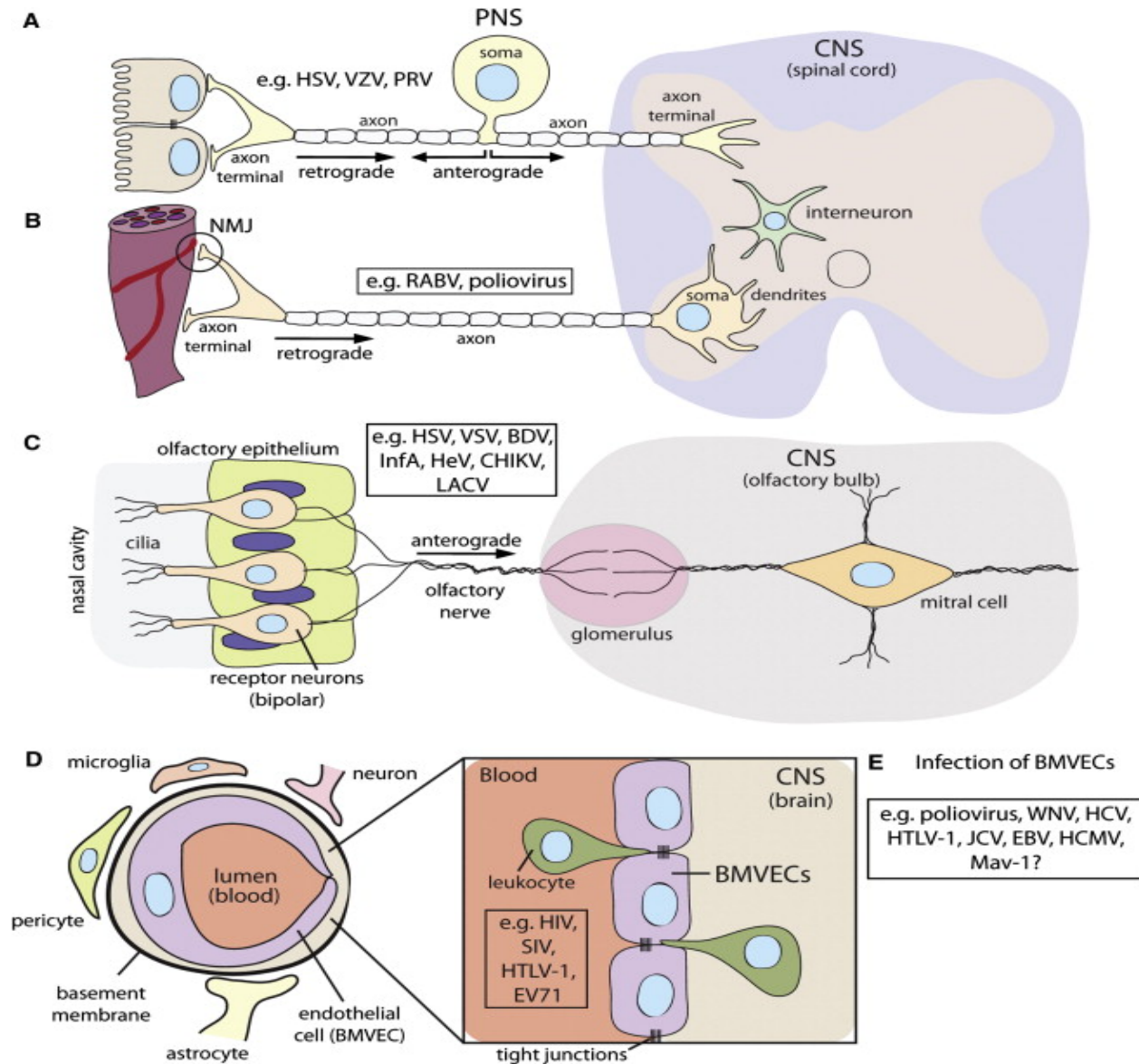
- By the end of this lecture, students should be able to:
 - 1. Identify infectious diseases affecting the peripheral nervous system (PNS).
 - 2. Explain the microbiology and pathogenesis of Varicella-Zoster Virus (VZV).
 3. Discuss clinical presentation, complications, and diagnosis of shingles
 4. Outline laboratory diagnosis methods for VZV.
 5. Summarize prevention and treatment strategies.

peripheral nervous system



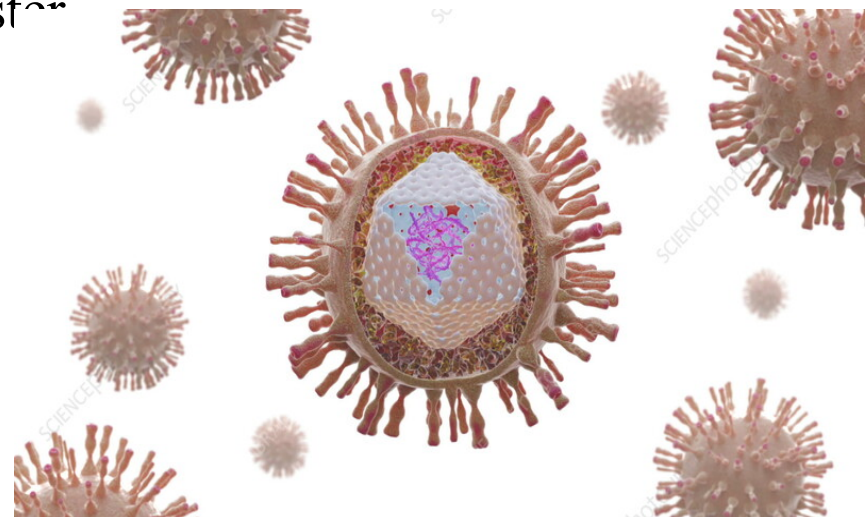
General Infectious Diseases Affecting the PNS

- **Leprosy** (*Mycobacterium leprae*) – chronic infection, peripheral neuropathy.
- **Poliomyelitis** (Poliovirus) – acute flaccid paralysis.
- **Diphtheritic neuropathy** (*Corynebacterium diphtheriae* toxin).
- **Lyme disease** (*Borrelia burgdorferi*) – radiculopathy, cranial neuropathy.
- **HIV**-associated neuropathy.
- **COVID-19** infection.
- Varicella-Zoster Virus (**Shingles**).

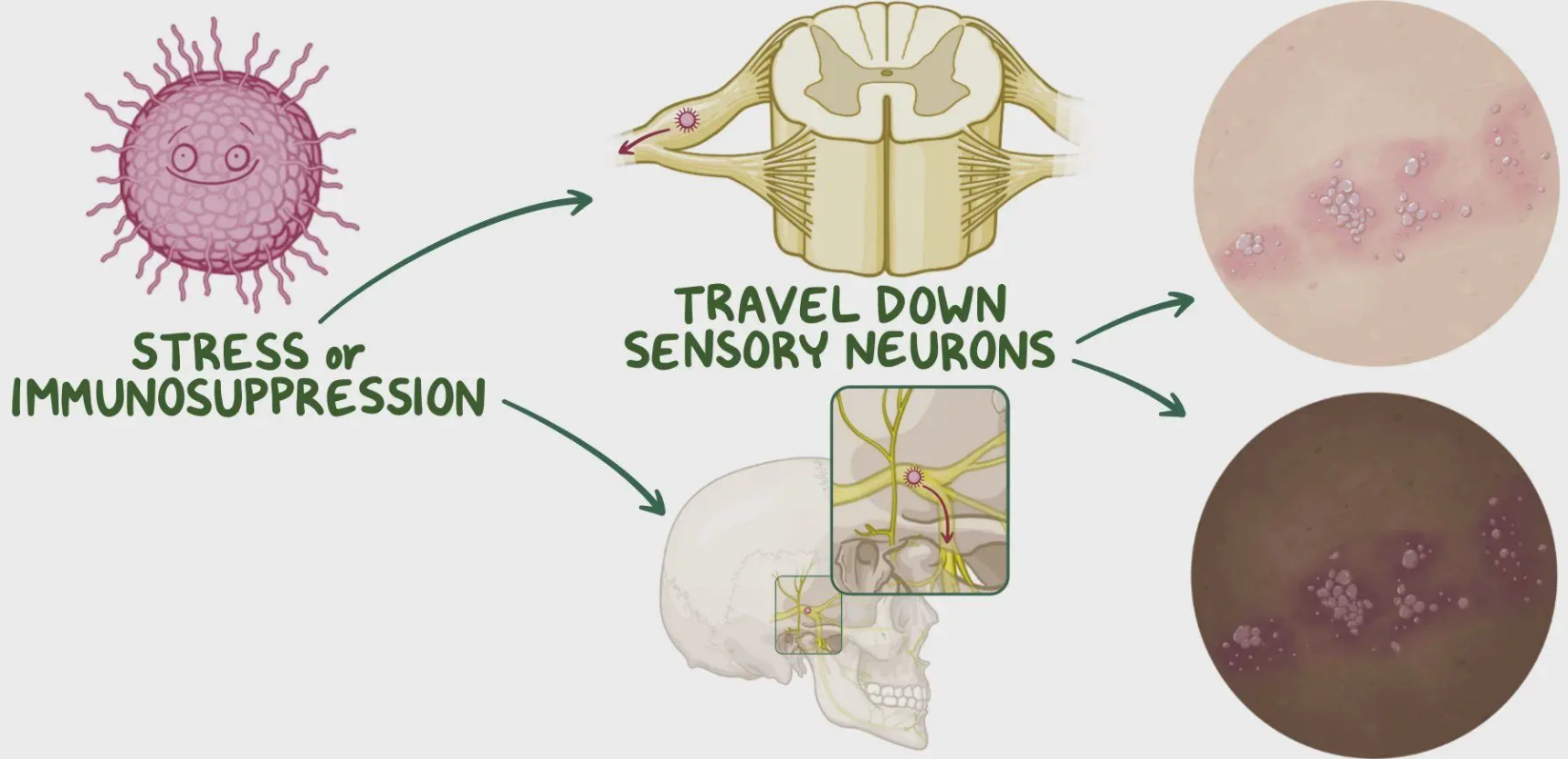


Shingles (Herpes Zoster)

- Caused by reactivation of **Varicella Zoster Virus** (VZV).
- After **primary infection (chickenpox)**, the virus remains latent in the dorsal root ganglia.
- Reactivation leads to a painful, vesicular rash along a dermatome.
- Common in elderly and immunocompromised patients.
- Major complications:
 - post-herpetic neuralgia
 - ophthalmic zoster
 - CNS involvement.



HERPES ZOSTER INFECTION (SHINGLES)



- **Classification:**

- - Family: Herpesviridae
- - Subfamily: Alphaherpesvirinae
- - Genus: Varicellovirus
- - DNA virus (linear, dsDNA, ~125 kb)

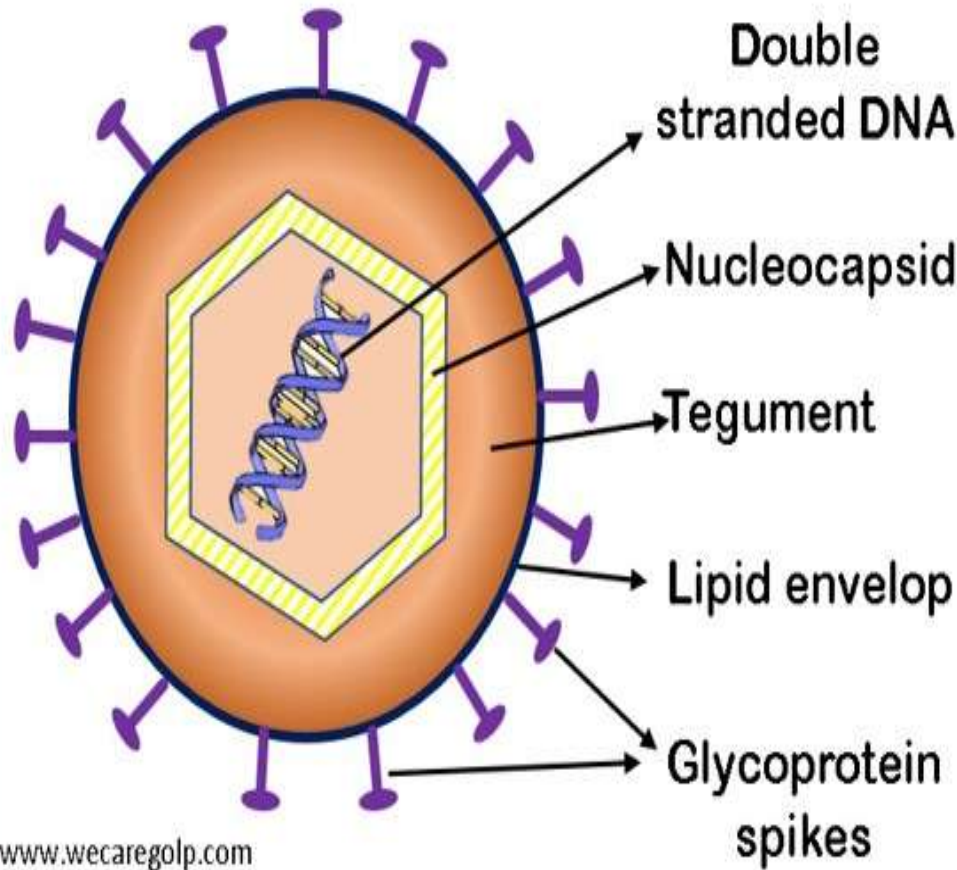
- **Structure:**

- - Icosahedral nucleocapsid
- - Tegument proteins
- - Enveloped with glycoprotein spikes

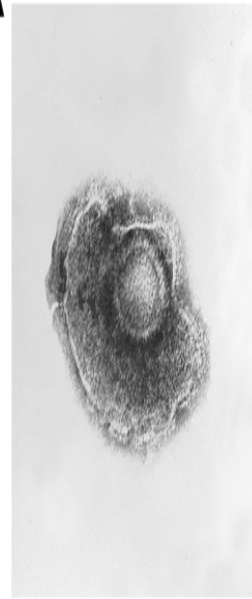
- **Replication:**

- - Nuclear replication
- - Latency in dorsal root & cranial nerve ganglia
- - Reactivation → Herpes Zoster (shingles)

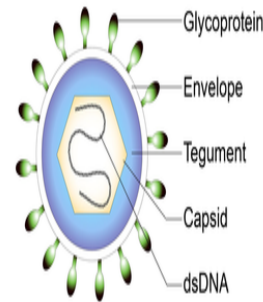
Varicella-Zoster Virus (VZV)



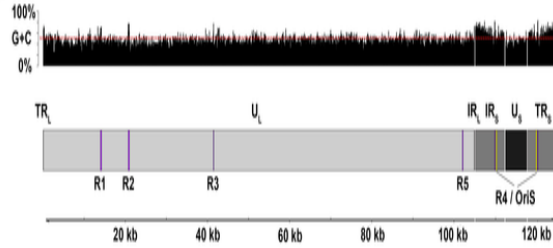
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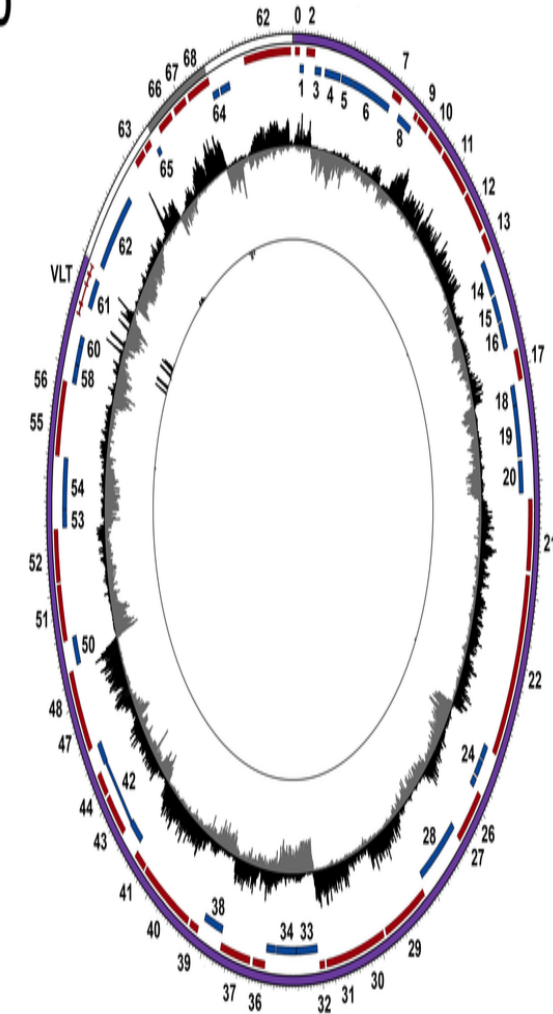
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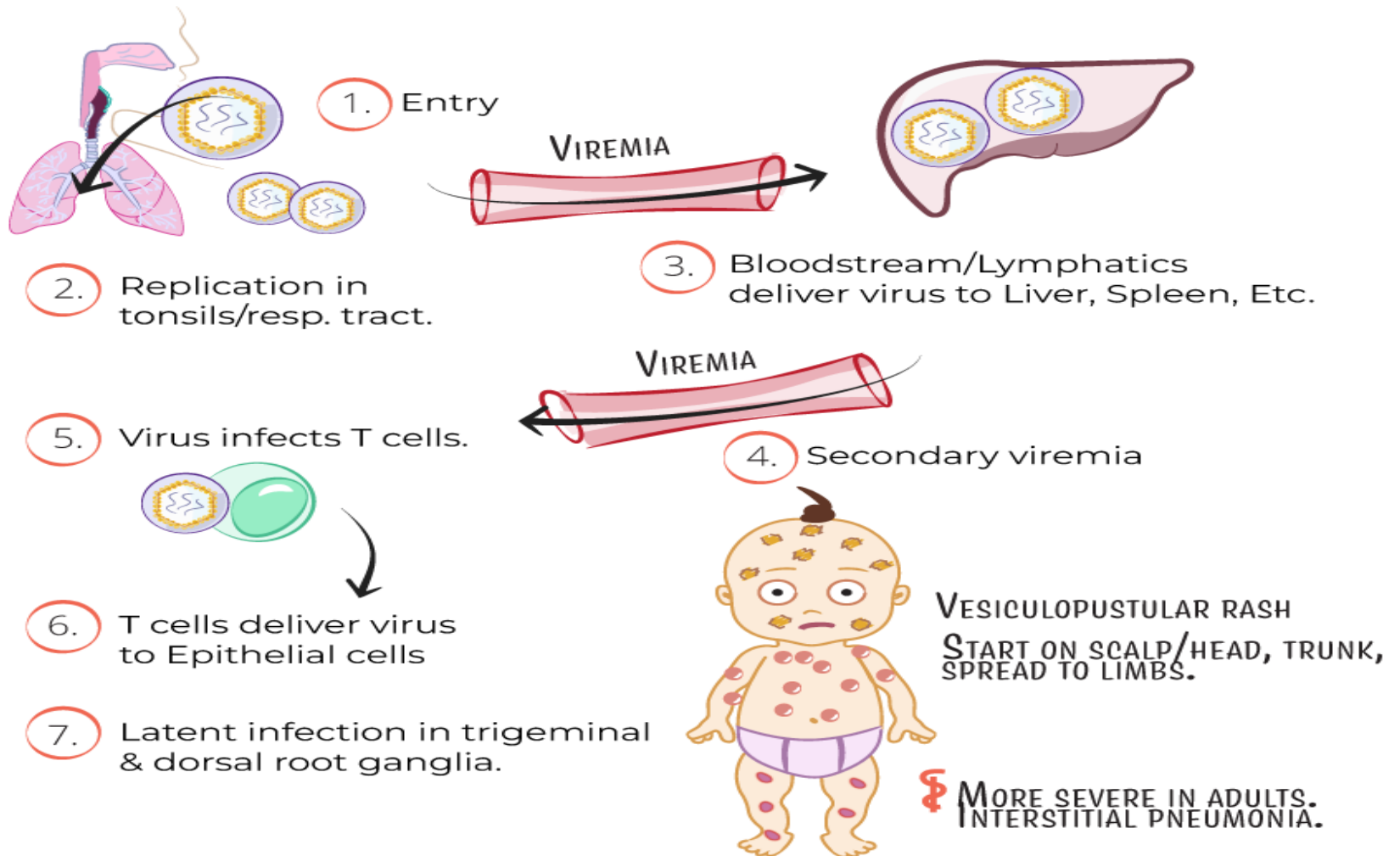


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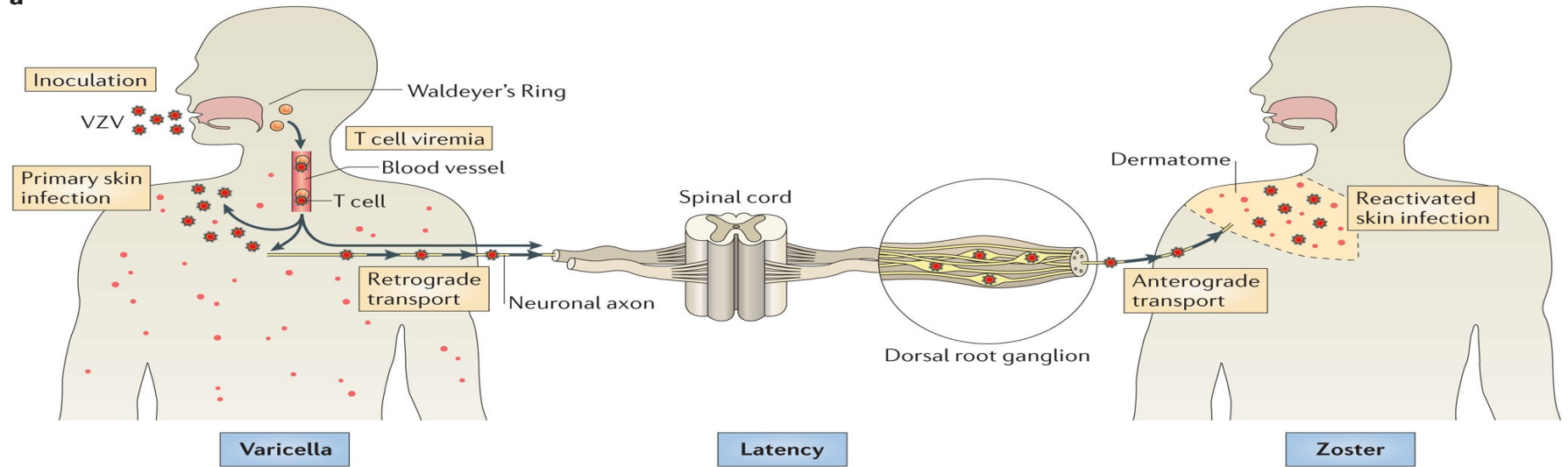
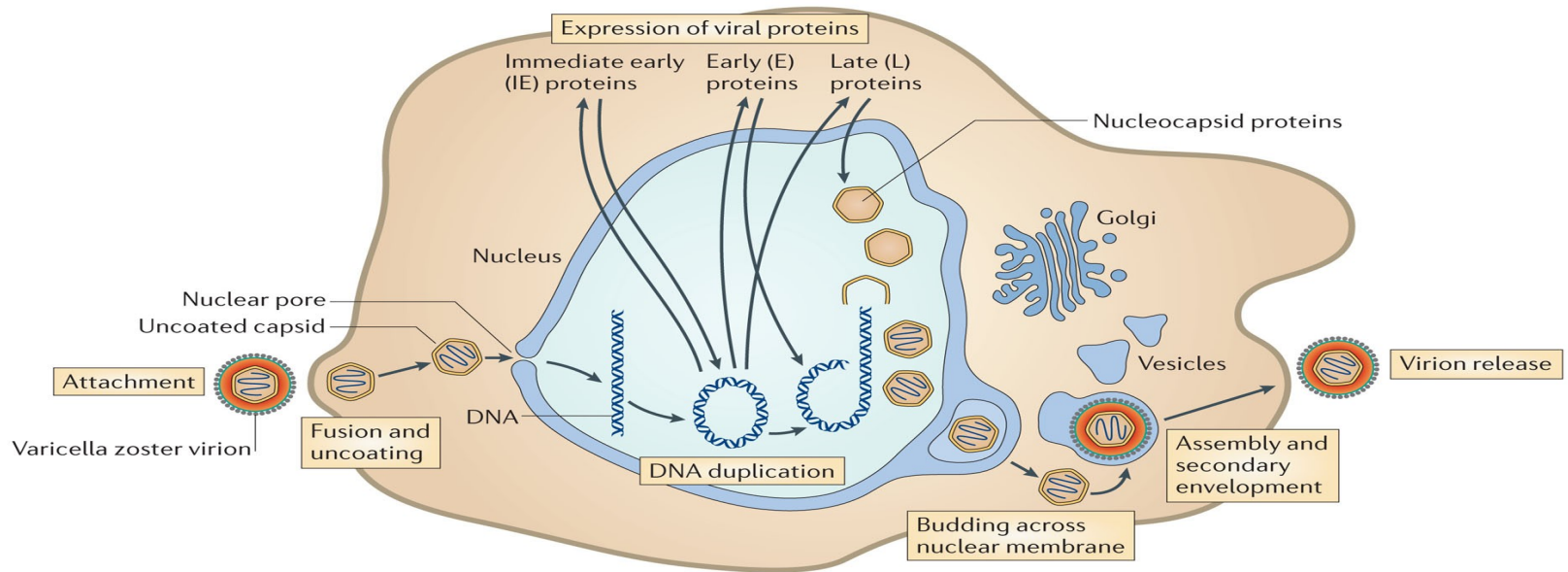
Varicella-Zoster Virus (aka, HHV-3)

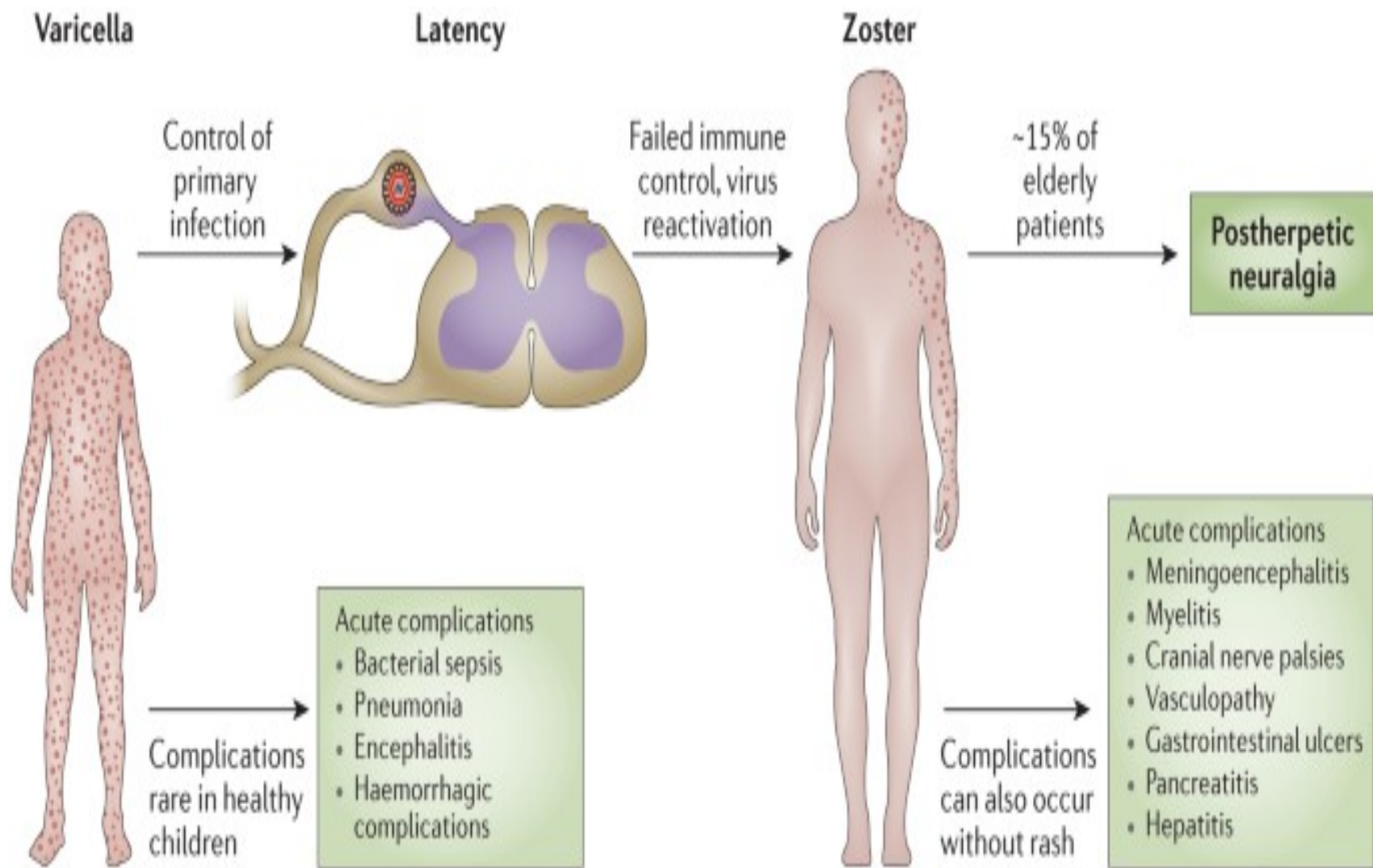
CHILDREN: MILD RESPIRATORY ILLNESS & SUCCESSIVE CROPS OF MACULOPAPULAR RASH.



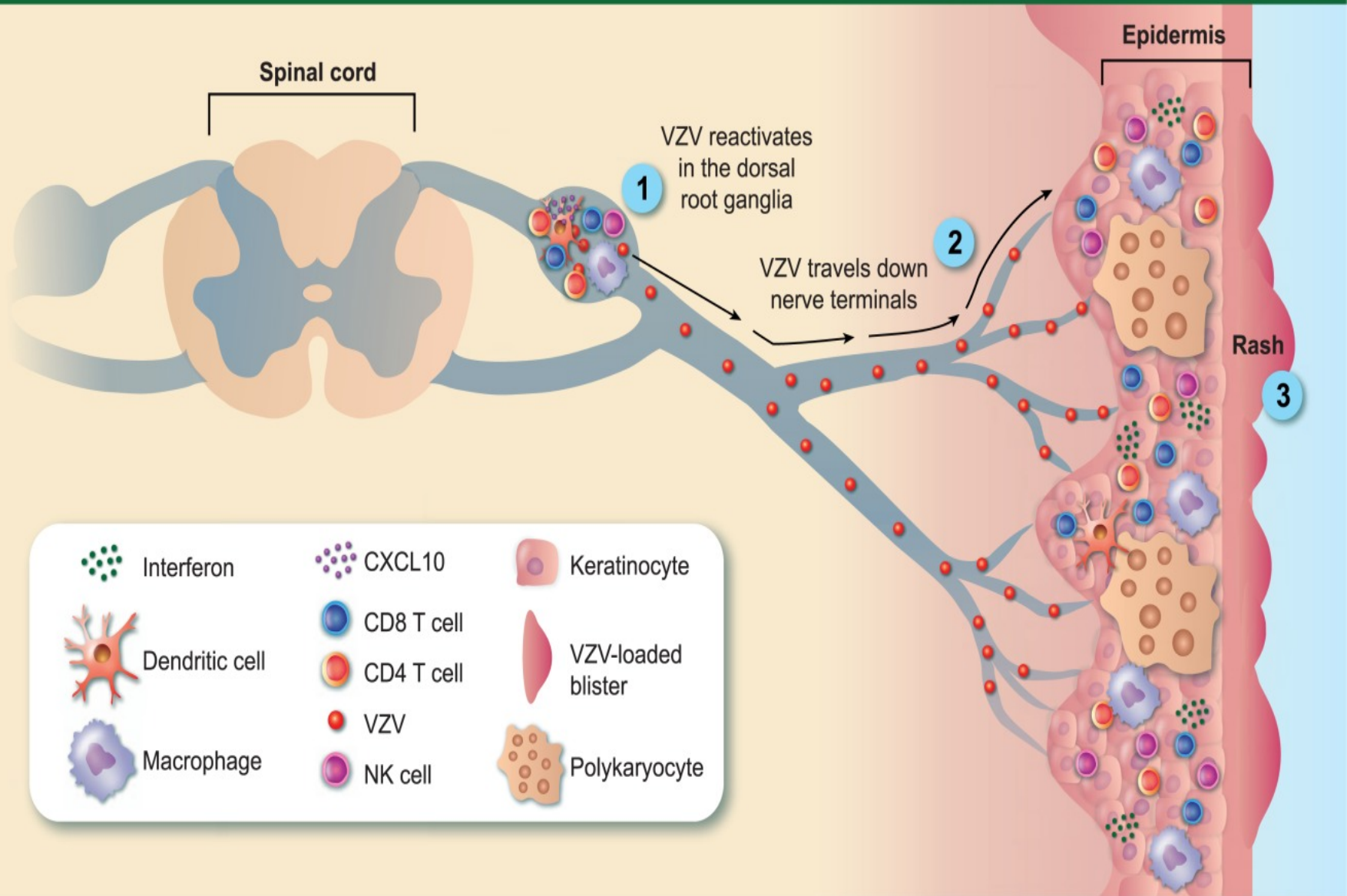
Pathogenesis of VZV

- 1. **Primary infection** → Chickenpox (varicella).
- 2. **Latency** → Virus resides in **sensory dorsal root ganglia**.
- 3. **Reactivation** → Triggered by immunosuppression, stress, or aging.
- 4. **Spread** → Virus travels along sensory nerves to the skin.
- 5. **Cytopathic Effect:**
 - Multinucleated giant cells
 - Cowdry type A intranuclear inclusions
- 6. **Clinical manifestation** → Painful vesicular rash in dermatomal distribution.

a**b**



VZV reactivation

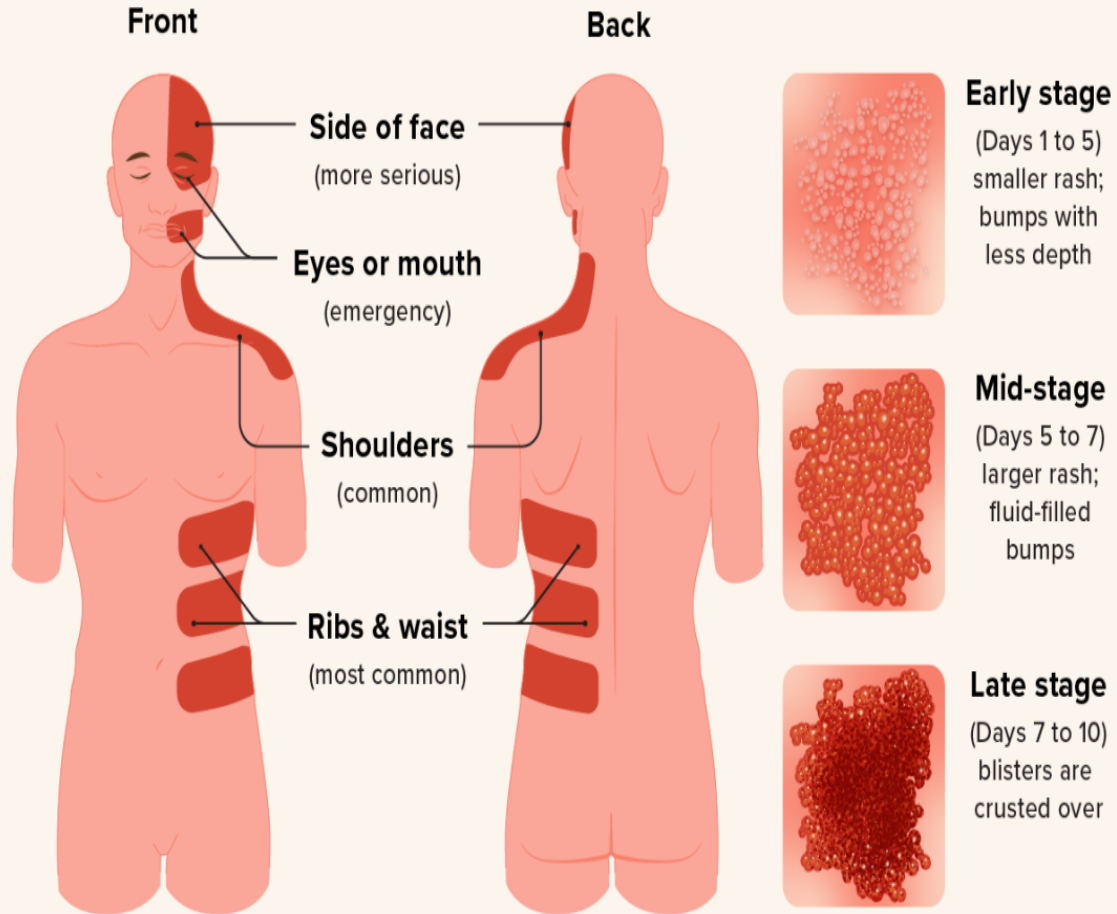


Clinical Features of Shingles

- Prodrome: Pain, burning, or tingling in the affected dermatome.
- Rash: Unilateral, vesicular eruption in dermatomal distribution.
- Systemic symptoms: Fever, malaise (sometimes).
- Complications:
 - - Post-herpetic neuralgia (chronic pain).
 - - Ophthalmic zoster → vision loss.
 - - Neurological: meningitis, encephalitis, myelitis.

Shingles Lesions

Common locations and stages



healthline



Shingles Diagnosis



physical exam



PCR



**serological
testing**



Laboratory Diagnosis of VZV

- **Clinical** diagnosis: typical rash distribution.
- **Tzanck smear**: multinucleated giant cells (low specificity).
- **PCR**: detects VZV DNA – gold standard.
- **Direct immunofluorescence** for viral antigens.
- **Serology**: VZV IgM (acute), IgG (immunity).

Table 2: Principles and methods of the laboratory diagnosis of varicella-zoster virus infections

Principle	Method	Material for analysis/Transportation	Note
Demonstration of viral DNA	Polymerase chain reaction (PCR)	Liquor, vesicle contents, tissue, broncho alveolar lavage (BAL), EDTA, blood, amniotic fluid	Method of choice
Demonstration of viral antigens	Indirect immunofluorescence	Vesicle contents rich in cells, tissue	Alternative method
Virus Isolation	Isolation in cell culture	Vesicle contents, tissue, BAL, EDTA blood, amniotic fluid/special medium for transportation required	Used when special investigations are required
Differentiation between wild- and vaccine-derived virus	PCR, Restriction fragment length polymorphism analysis, sequencing	Viral isolates, vesicle contents, tissue	Performed in specialized laboratories
Determination of resistance against antiviral drugs	Plaque reduction assay	Viral isolates	Performed in specialized laboratories
Demonstration of antibodies	Immunofluorescence, quantitative immunoassays	Serum, liquor	Determination of immune status, retrospective confirmation of varicella or zoster

Prevention and Treatment

Prevention:

- Varicella vaccine (childhood).
- Zoster vaccine (older adults, immunocompromised).

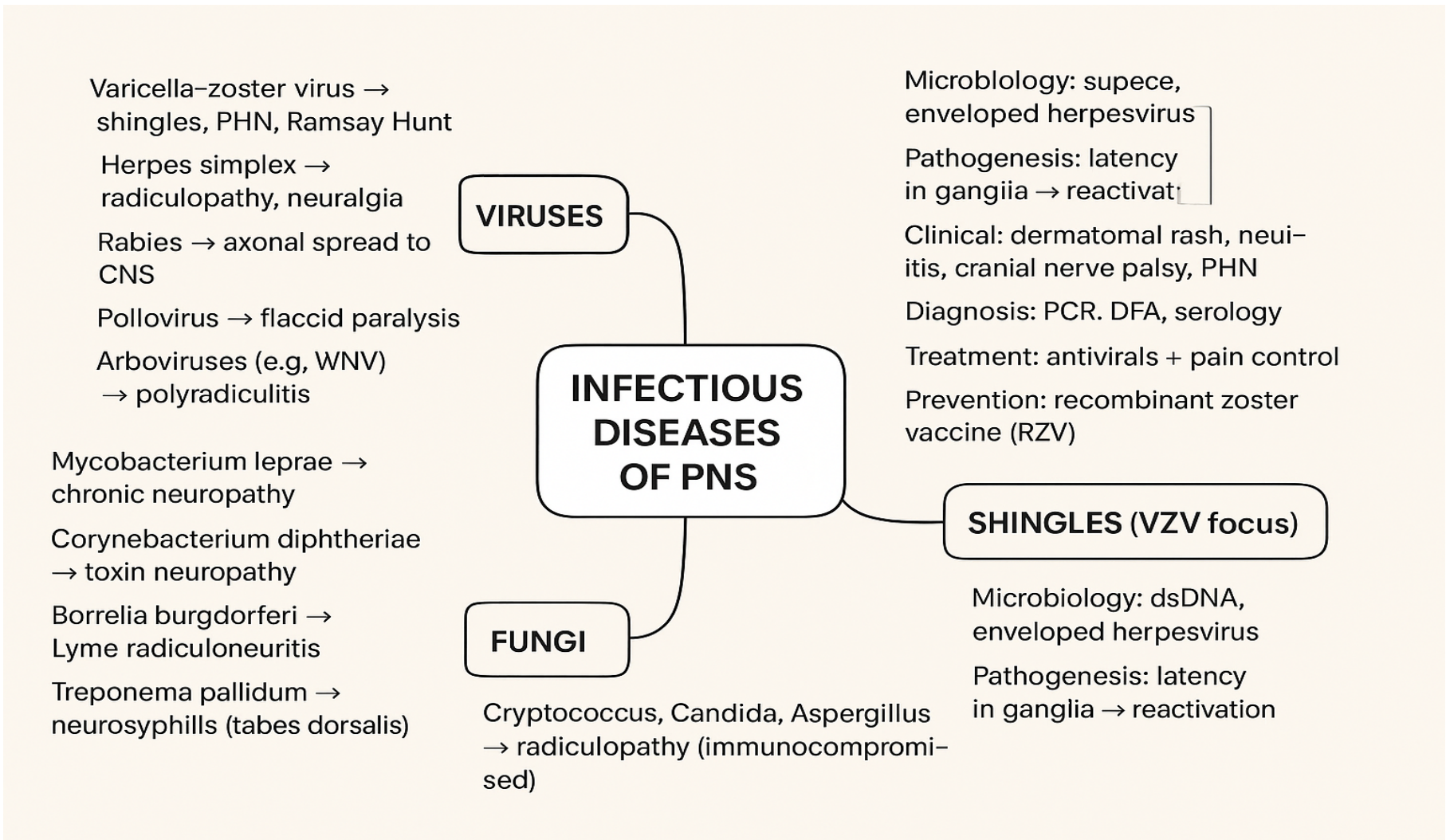
Treatment:

- Antivirals: Acyclovir, Valacyclovir, Famciclovir (early treatment reduces severity).
- Pain control: NSAIDs, opioids, gabapentin.
- Corticosteroids: sometimes used to reduce inflammation.
- Management of complications (ophthalmic zoster → urgent referral).



- A 65-year-old man presents with severe burning pain on the right side of his chest, followed by the appearance of a unilateral vesicular rash along the T5 dermatome. He had chickenpox during childhood.
- What is the most likely diagnosis?
- A) Herpes simplex virus infection
- B) Varicella (chickenpox)
- C) Herpes zoster (shingles)
- D) Contact dermatitis

Summary



Herpes Zoster - Shingles: *Pathogenesis and clinical findings*

Authors:

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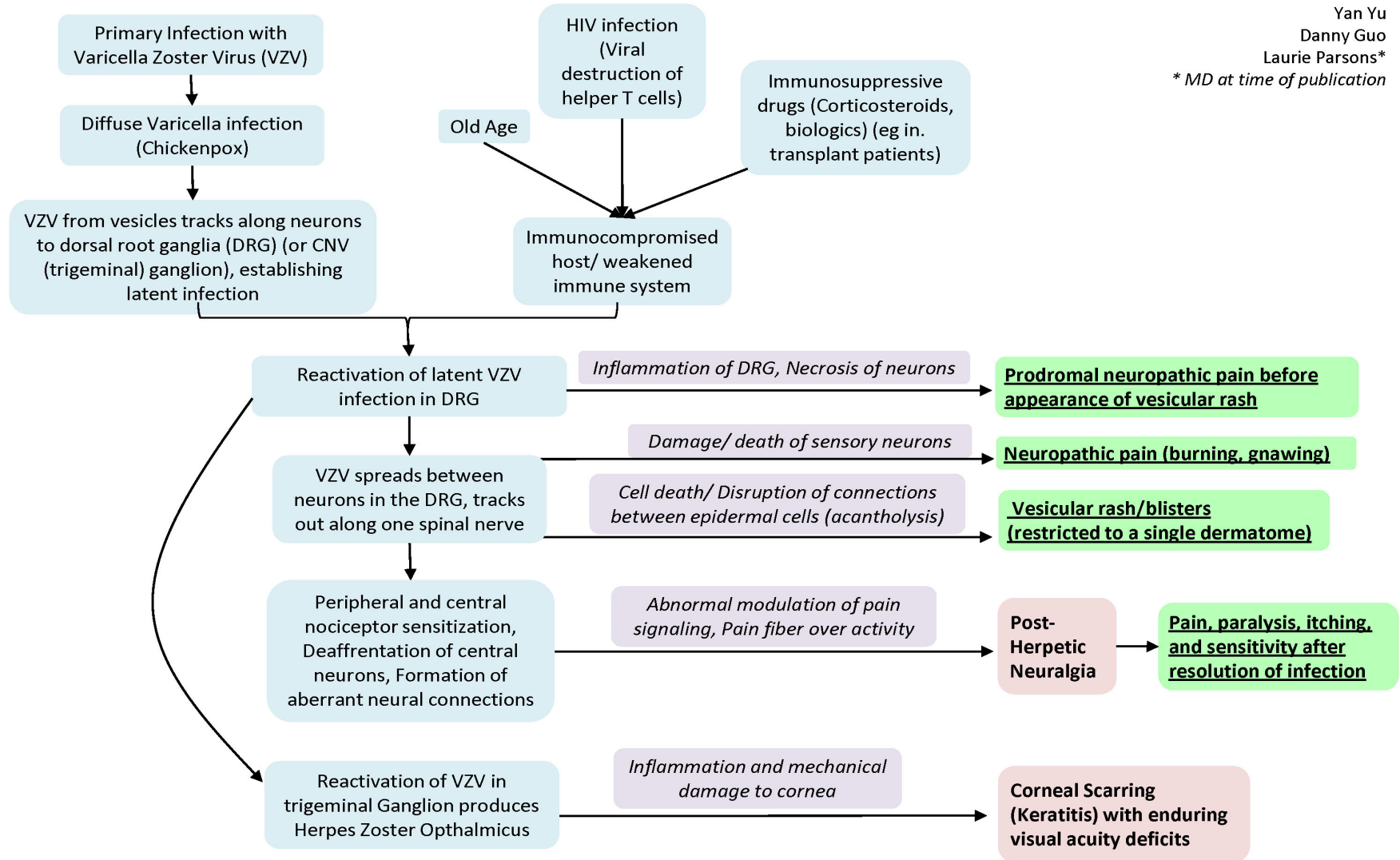
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Legend: Pathophysiology Mechanism Sign/Symptom/Lab Finding Complications

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- 4. CDC – Varicella and Shingles information (<https://www.cdc.gov/shingles/>).
- 5. WHO – Varicella and Herpes Zoster Fact Sheets (<https://www.who.int/>).



Any
Question?

